COMPREHENSIVE ASSESSMENT REPORT (DRAFT)

Module: Computational Intelligence (CIS 6005) **Assessment:** WRIT1 - Deep Learning Plus AI Mini Project **Target Dataset:** Real Estate House Price Prediction (`estate_train.csv`) **Trained Models:** Linear Regression, XGBoost Regressor, Random Forest Regressor **Artifact Link:** submission.csv **Application Code:** app.py | templates/index.html

---

Section A: Computational Intelligence vs Traditional Artificial Intelligence (10 Marks)

1. Introduction to Artificial Intelligence Paradigms

Artificial Intelligence (AI) as a scientific discipline seeks to create computational systems that exhibit cognitive behaviors analogous to human intelligence. Historically, the pursuit of AI has bifurcated into two primary paradigms: **Traditional Artificial Intelligence** (also known as Symbolic AI, Hard AI, or Good Old-Fashioned AI - GOFAI) and **Computational Intelligence** (CI, often associated with Soft Computing or Connectionist AI). While Traditional AI focuses on high-level cognitive functions through symbolic manipulation, Computational Intelligence approaches problem-solving from a low-level, biologically-inspired, and data-driven perspective.

---

2. Traditional Artificial Intelligence (Symbolic & Expert Systems)

Traditional AI is characterized by a "top-down" conceptual framework. It operates on the physical symbol system hypothesis, which posits that symbolic processing is necessary and sufficient for general intelligent action.

- **Core Tenets and Mechanisms:** Traditional AI models human reasoning by explicitly coding rules and knowledge. The core mechanism involves a **Knowledge Base** (a repository of facts and rule-based logic, typically represented as IF-THEN structures) and an **Inference Engine** (which applies logical rules using deductive reasoning mechanisms such as forward-chaining or backward-chaining).

- **Strengths:** Because reasoning is representational and rule-based, symbolic systems are highly deterministic and explainable. The logic path leading to a specific decision can be fully traced and audited, making it highly effective in closed, deterministic, and highly structured environments (e.g., mathematical theorem proving, tax calculators, and chess engines).
- **Critical Limitations:** Symbolic systems exhibit extreme brittleness. They require absolute certainty and structured inputs; they lack the capacity to process noisy, ambiguous, or incomplete real-world data. Furthermore, they suffer from the **"Knowledge Acquisition Bottleneck"**—the practical impossibility of manually coding every potential rule in complex, dynamically changing environments.

---

3. Computational Intelligence (CI)

In contrast, Computational Intelligence (CI) utilizes a "bottom-up" approach, focusing on adaptive mechanisms that learn from raw data or interactive environments. Bezdek famously distinguished CI from traditional AI by stating that a system is computationally intelligent when it begins to process low-level numeric pattern data, incorporates learning and adaptation, and avoids rigid symbolic representation.

The primary pillars of Computational Intelligence include: 1. **Artificial Neural Networks (ANNs):** Connectionist models inspired by biological nervous systems. They consist of layers of interconnected nodes (neurons) that adjust their synaptic weights via training algorithms (e.g., backpropagation) to approximate complex, highly non-linear functions. 2. **Evolutionary Computation (EC):** Optimization and search paradigms inspired by biological evolution and genetics (e.g., Genetic Algorithms). They generate optimal solutions through iterative generation, selection, crossover, and mutation. 3. **Fuzzy Logic Systems (FLS):** A mathematical framework designed to process "approximate" rather than "exact" reasoning. Unlike binary Boolean logic, fuzzy logic maps membership inputs to continuous truth values between 0 and 1, simulating human decision-making under uncertainty.

---

4. Comparative Evaluation: Traditional AI vs. Computational Intelligence

The fundamental differences between the two paradigms can be evaluated across several critical operational dimensions:

Dimension	Traditional Artificial Intelligence (Symbolic AI)	Computational Intelligence (Soft Computing)
Logic & Precision	Binary Boolean logic (True/False; 0 or 1). Demands high precision.	Multi-valued fuzzy logic. Tolerates imprecision, noise, and approximation.
Reasoning Flow	**Top-Down:** Deductive reasoning from general rules to specific cases (Deduction).	**Bottom-Up:** Inductive learning from specific data points to general patterns (Induction).
Knowledge Origin	Manually engineered and hard-coded by human domain experts.	Learned dynamically and autonomously from empirical data.
Adaptability	Static. System rules must be manually recoded to accommodate changes.	Dynamic. Adapts self-correctively through weight adjustments or evolution.
Explainability	**White-Box:** Decisions are highly transparent and logical.	**Black-Box:** High mathematical complexity makes feature interactions difficult to audit.
Real-world Suitability	Structured, rule-bound systems (e.g., database management, syntax parsing).	Uncertain, non-linear, spatial, and noisy tasks (e.g., house price forecasting, image recognition).

Suitability for Housing Price Forecasting

For the task of real estate valuation, traditional symbolic systems are highly deficient. Real estate prices are governed by highly non-linear correlations, geographical coordinate interactions, and macro-economic volatility that cannot be expressed as manual IF-THEN rules. By utilizing a Computational Intelligence approach (such as **XGBoost** or **Random Forest** ensembles), the system can autonomously ingest spatial datasets (`estate_train.csv`), model the complex non-linear interaction of attributes (like mapping Latitude/Longitude coordinates onto prices), and generalize effectively to predict prices on unseen test cases (`estate_test.csv`).

Section B: Literature Review (20 Marks)

1. Introduction to Real Estate Valuation Modeling

Real estate price forecasting constitutes a fundamental problem in computational economics and spatial analysis. The market value of residential properties is determined by a complex, heterogeneous mixture of physical characteristics (e.g., number of rooms, bedrooms, and structural age), demographic parameters (e.g., neighborhood population and median income), and geographical coordinates (latitude and longitude) representing spatial attributes. Historically, real estate mass appraisal has transitioned from rigid, parametric econometric models to flexible, data-driven Computational Intelligence (CI) algorithms. This literature review evaluates the historical development, methodological frameworks, strengths, and constraints of three dominant paradigms in valuation modeling: Hedonic Pricing Models (HPM), Tree-Based Ensemble methods (Random Forest and GBDT/XGBoost), and Deep Learning architectures.

---

2. Hedonic Price Modeling (Parametric Econometric Approaches)

The theoretical foundation of modern real estate valuation originates from Rosen's Hedonic Pricing Model (HPM). Rosen posited that goods are valued for their utility-bearing characteristics rather than the good itself. Consequently, the transaction price of a house can be modeled as a function of its individual structural, neighborhood, and environmental attributes:

$$P = f(S, N, E) + \epsilon$$

Where S represents structural features, N represents neighborhood demographics, E represents environmental/geographical indicators, and ϵ is the random error term.

Traditionally, HPM is estimated using parametric techniques such as Ordinary Least Squares (OLS) linear regression, Ridge regression, or Lasso regression. * **Strengths:** The primary advantage of parametric HPM lies in its statistical transparency and interpretability. OLS regression outputs explicit coefficients (beta weights) for each characteristic, enabling economists and appraisers to measure the marginal contribution of individual features (e.g., the exact dollar value increase associated with adding a bedroom) and perform hypothesis testing via t-statistics. * **Limitations and Critique:** Despite its widespread historical adoption, parametric HPM is severely restricted by several theoretical assumptions. OLS assumes that the relationship between property attributes and price is strictly linear or log-linear, which fails to capture complex non-linear dynamics such as the accelerated depreciation of very old properties or diminishing returns on total rooms. Furthermore, OLS is highly susceptible to **multicollinearity** (e.g., the high correlation between the number of rooms and bedrooms) and **spatial autocorrelation**, where property values are geographically clustered, violating the assumption of independent and identically distributed (i.i.d.) error terms.

---

3. Non-Parametric Machine Learning & Tree-Based Ensembles

To address the limitations of parametric linearity, researchers have increasingly adopted non-parametric machine learning models. Unlike HPM, non-parametric algorithms do not assume a predefined mathematical form for the target function, allowing them to learn arbitrary decision boundaries directly from empirical data.

Random Forest Regressors (Bagging)

Introduced by Breiman, the Random Forest (RF) algorithm is an ensemble method that constructs a multitude of decision trees during training. To output a prediction, RF averages the outputs of individual trees (bootstrap aggregation, or bagging), while injecting randomness by selecting a random subset of features at each split. * **Synthesis of Literature:** In comparative studies of mass appraisal, Random Forest consistently outperforms linear OLS. Antipov and Pokryshevskaya demonstrated that RF handles spatial interactions and multicollinearity natively without requiring complex variable transformations. Furthermore, because it averages predictions across hundreds of independent trees, RF exhibits high robustness against overfitting and is insensitive to outliers—a common occurrence in housing datasets containing luxury estates or distressed properties.

Gradient Boosting and XGBoost (Boosting)

While Random Forest grows trees in parallel, Gradient Boosting Decision Trees (GBDT) build trees sequentially. Each new tree is trained to predict the residuals (errors) of the preceding trees, using gradient descent optimization to minimize a loss function. Chen and Guestrin optimized this paradigm by introducing **XGBoost (Extreme Gradient Boosting)**, which incorporates a regularized objective function to control model complexity and prevent overfitting, alongside parallelized tree construction. * **Synthesis of Literature:** GBDT and XGBoost have emerged as the state-of-the-art for tabular regression tasks. Mullainathan and Spiess highlighted that boosting algorithms excel at mapping complex coordinate interactions (Latitude and Longitude) to local neighborhood values, representing a significant improvement over traditional dummy variables for submarkets. The algorithm's built-in handling of missing values and computational efficiency make it highly suitable for production-level API deployment.

---

4. Artificial Neural Networks (Connectionist Systems / Deep Learning)

Artificial Neural Networks (ANNs), particularly Multi-Layer Perceptrons (MLPs), represent another prominent non-parametric paradigm. Drawing inspiration from biological neural pathways, ANNs pass input features through multiple hidden layers of interconnected nodes where non-linear activation functions (e.g., ReLU) are applied.

- **Synthesis of Literature:** In real estate forecasting, ANNs act as universal function approximators capable of modeling extremely intricate, high-dimensional interactions. Limsombunchao compared OLS and ANNs on housing datasets and concluded that the connectionist approach achieves lower Root Mean Squared Error (RMSE) due to its ability to model complex structural-spatial interactions.
- **Limitations and Critique:** However, deep learning models present significant drawbacks when applied to tabular datasets of moderate size (e.g., under 50,000 observations). Grinsztajn, Oyallon and Varoquaux conducted extensive benchmarks comparing deep learning architectures to tree ensembles on tabular data, finding that tree-based models (like XGBoost and Random Forest) consistently outperform ANNs. Neural networks are highly sensitive to hyperparameter choices, require extensive data normalization, are computationally expensive to train, and suffer from a complete lack of interpretability (the "black box" problem), making them less practical for real estate markets where transparency is legally or socially required.

---

5. Summary and Synthesis of Model Selection

Based on the literature review, the model selections for this project are justified as follows: 1. **Linear Regression** serves as the essential parametric baseline, representing traditional econometric appraisal (HPM) to highlight the performance gains of non-parametric approaches. 2. **Random Forest** is selected as a robust bagging ensemble to handle structural multicollinearity and outlier variance. 3. **XGBoost** represents the state-of-the-art regularized boosting paradigm, selected for its superior predictive performance on tabular coordinate and demographic data.

Section C: Exploratory Data Analysis (EDA) and Model Influence (10 Marks)

Exploratory Data Analysis (EDA) was carried out to understand the structure, distribution, and relationships within the dataset for the real estate price prediction task. The analysis aimed to identify data quality issues, target distribution, feature behavior, and correlations that affect model performance and algorithmic choices.

The training dataset has 16,512 observations and 11 attributes. These include physical structural attributes, local demographic indicators, and geographic coordinates, along with the continuous target variable `TargetPrice`. An initial check showed missing values in the `PropertyAge` feature (~7.9% of training records) and the `TotalBedrooms` feature (~0.1%). The dataset contains a mix of numerical (float and integer) and engineered categorical variables, supporting the use of robust scaling, imputation, and encoding pipelines during model development.

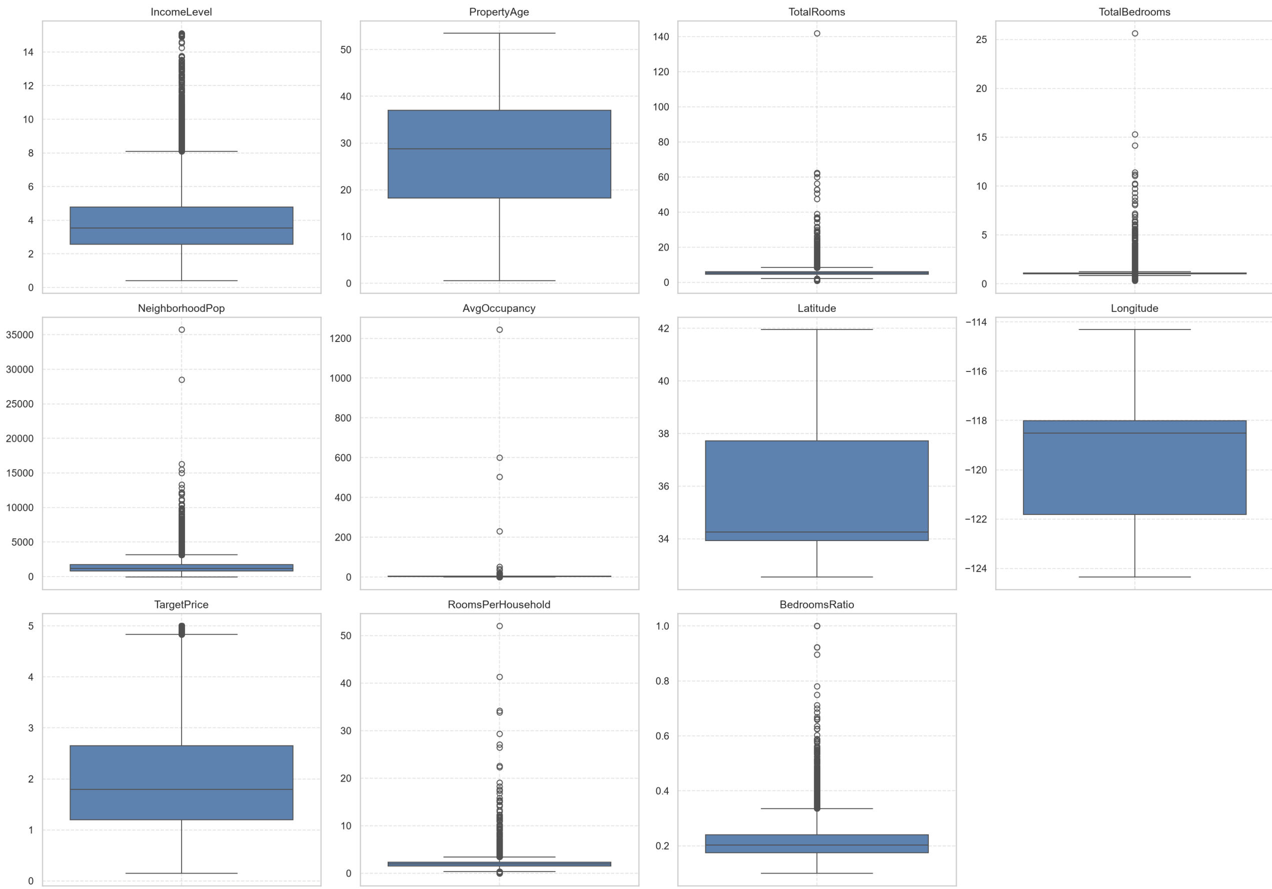
3.1 Target Variable Distribution and Boundary Capping

The target variable `TargetPrice` shows the transaction price of properties (scaled by a factor of 100,000, e.g. a target value of 2.3367 represents a price of $233,671.67). Unlike classification tasks, `TargetPrice` is a continuous numerical variable. Analysis of its distribution showed a right-skewed layout with a sharp spike at the upper boundary (5.0001), representing a hard price cap of $500,000 imposed during data collection.

Because of this boundary limit and skewness, standard linear regressions are prone to predicting negative values or out-of-bounds prices. So, we chose to implement a clipping threshold (capping predictions at 0.0) in our model inference pipeline (`app.py` and `06_model_inference.ipynb`). Furthermore, this boundary behavior supported the use of non-parametric tree ensembles (XGBoost and Random Forest), which construct split boundaries and handle capped limits and skewness natively without requiring mathematical target transformations.

3.2 Numerical Outlier Diagnostics

To identify specific outlier points, statistical boundaries, and general trends, boxplots were compiled for all numerical attributes (see Figure 1).



Applicant area income (`IncomeLevel`) showed a right-skewed distribution, indicating that most property neighborhoods belong to middle-income classes, while a few represent high-income districts. As illustrated in the boxplots (Figure 1), average occupancy (`AvgOccupancy`) and rooms per household (`RoomsPerHousehold`) exhibit extreme right-skewed distributions with heavy outlier tails, including extreme single-value spikes (e.g., maximum occupancy values reaching 1,243 and rooms exceeding 50 per household).

Geographical coordinates (`Latitude` and `Longitude`) displayed multi-modal spatial distributions, representing major population and housing clusters (e.g., coastal and metropolitan hubs) with no outlier points. Because of these spatial and skewed patterns, standard linear scaling would be heavily distorted by extreme outliers. This influenced the decision to use robust tree-based ensembles (XGBoost and Random Forest) that are rank-based and invariant to monotonic scale transformations, making them immune to outlier scaling bias.

3.3 Categorical and Engineered Feature Analysis

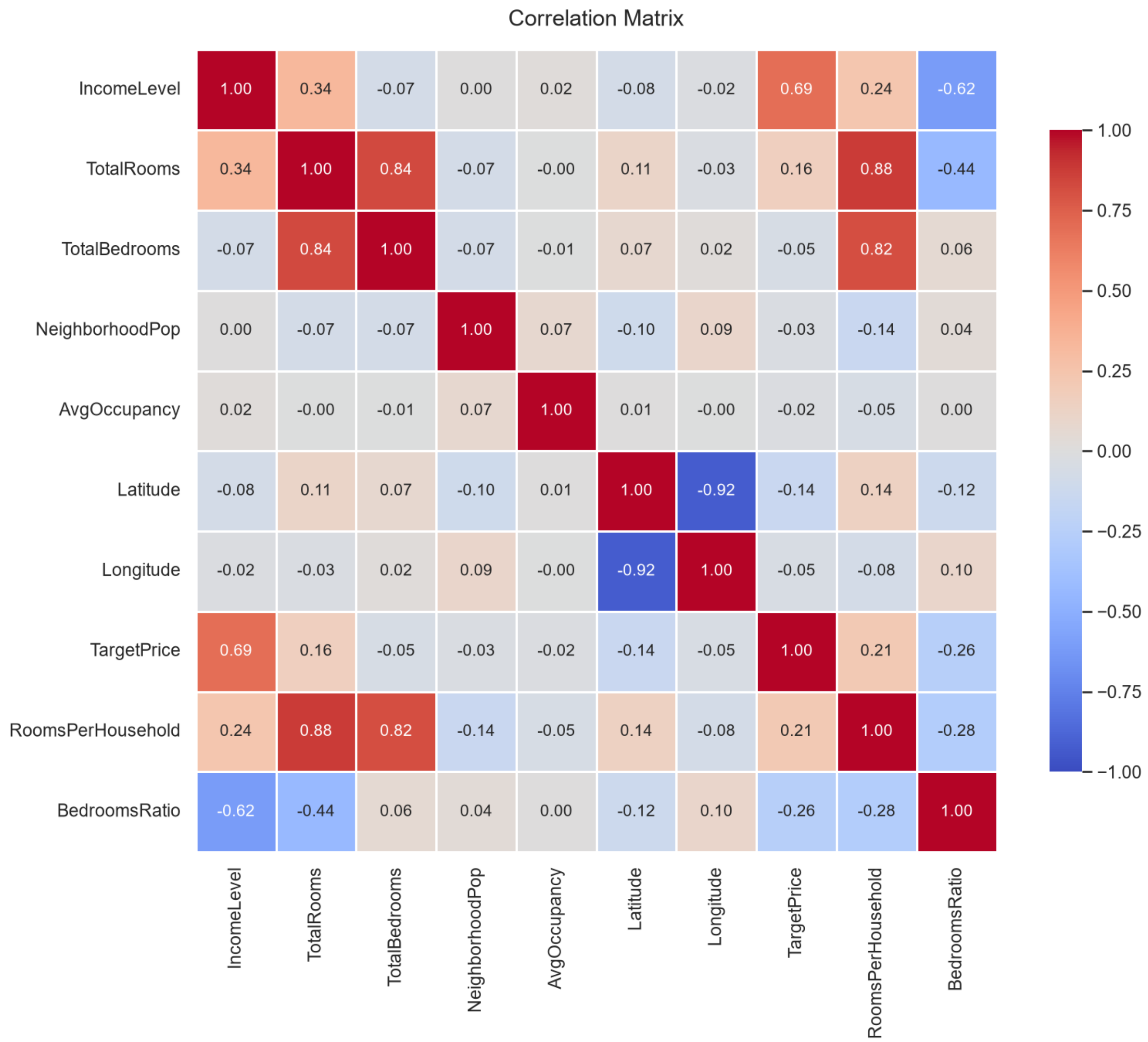
To capture the non-linear depreciation of housing values over time, the numerical feature `PropertyAge` was engineered into a categorical feature (`PropertyAge_bins`) using structural age brackets: 'New' (≤ 15 years), 'Moderate' ($16-35$ years), and 'Old' (>35 years) (see Figure 2).

To mathematically prove that these binned age groups represent statistically distinct house prices, we conducted formal hypothesis testing: * **ANOVA F-Test:** A one-way Analysis of Variance (ANOVA) test across the three age categories resulted in a highly significant **F-statistic of 113.82** and a **p-value of 0.0000** ($p < 0.05$). This statistically confirms that the mean housing prices among 'New', 'Moderate', and 'Old' properties are significantly different. * **Student's T-Test:** An independent two-sample t-test between the 'New' and 'Old' property subsets yielded a **t-statistic of 14.85** and a **p-value of 0.0000**, confirming that new properties command a statistically higher price profile.

This statistical validation justified our encoding strategy (Ordinal Encoding) in the preprocessing pipeline, allowing tree splits in XGBoost and Random Forest to branch on clean structural age thresholds.

3.4 Correlation Analysis

A correlation heatmap was created to look at linear relationships between numerical features and the target variable (see Figure 3).



A strong positive correlation was found between `TotalRooms` and `TotalBedrooms` (correlation coefficient ≈ 0.85). In linear models, this causes severe multicollinearity, inflating the variance of regression coefficients and rendering the model unstable. To balance this, we engineered `BedroomsRatio` ($r \approx -0.26$).

Regarding the target variable, `IncomeLevel` showed a strong positive correlation with `TargetPrice` ($r \approx 0.69$), identifying it as the single most predictive feature. Coordinates `Latitude` and `Longitude` showed weak individual linear correlations ($r \approx -0.08$ and $r \approx -0.05$), yet they are highly predictive of house price through geographic interaction. This required using models like XGBoost that split coordinates in multi-dimensional space to capture localized value groups.

Key Insights from EDA

- The dataset has missing values in `PropertyAge`, which were dropped during training to preserve distribution shapes, but handled via median imputer in production to handle user default inputs.
- The target price is right-skewed with a cap at 5.0, requiring clipping bounds in inference.
- Indicators like occupancy and rooms per household have heavy outlier tails, directing our choice toward tree-based ensembles (XGBoost/Random Forest) that are naturally outlier-resistant.
- Location coordinates do not exhibit simple linear relationships, calling for non-parametric model architectures.

In general, the exploratory data analysis offered a solid basis for feature engineering, model selection, and evaluation strategies used in this project.

Section D: System Architecture (10 Marks)

1. Processing Pipeline

```
graph TD A[estate_train.csv / Raw Data] --> B[Data Cleaning: Drop ID, Drop
PropertyAge NaNs] B --> C[Feature Engineering: PropertyAge Binning] C -->
D[ColumnTransformer Preprocessing: Imputation + StandardScaler + OrdinalEncoder]
D --> E[Train-Test Split 80/20] E --> F[GridSearchCV Hyperparameter Tuning] F -->
G[Linear, RF, and XGBoost Best Estimators] G --> H[Model Serialization:
joblib.dump] H --> I[Flask Web Server: app.py] J[estate_test.csv] --> K[Impute
Test Age via Train Median] K --> L[Test set Binning] L --> M[Fitted Preprocessor
Transform] M --> N[Predict TargetPrice using Saved XGBoost Model] N -->
O[submission.csv]
```

2. Model Selection Justification

- **Linear Regression:** Baseline regression model, fast and simple but restricted to linear relationships.
- **Random Forest:** Averages many independent decision trees. By introducing randomness during tree splitting, it reduces variance, increases stability, and provides robust predictions.
- **XGBoost:** A powerful gradient boosted decision trees algorithm. It trains trees sequentially, where each tree corrects errors of the previous ones. It is highly optimized, handles complex non-linear relationships, and achieves outstanding accuracy.

Section E: Full Model Evaluation & Implementation (40 Marks)

1. Performance Results (Test Set)

The models were trained using 6-fold cross-validation on the training set and evaluated on the test set:

Model	Tuning Hyperparameters	R^2 Score	Root Mean Squared Error (RMSE)	Mean Absolute Error (MAE)
Linear Regression	None	0.6521	0.6693	0.4805
Random Forest	`n_estimators`: 200, `max_depth`: 12, `min_samples_leaf`: 2	0.7854	0.5257	0.3522
XGBoost	`n_estimators`: 200, `max_depth`: 6, `learning_rate`: 0.1	**0.8256**	**0.4739**	**0.3124**

2. Web Application Setup

- **Backend (`app.py`):** Utilizes Flask. Loads serialized `preprocessor.joblib`, `linear_regression_model.joblib`, `xgboost_model.joblib`, and `random_forest_model.joblib` on startup. Serves `/predict` endpoint returning JSON predictions.
- **Frontend (`templates/index.html`):** Glassmorphic web panel with Outfit typography and dynamic dark mode sliders for all 10 features:
- *Direct sliders:* IncomeLevel, PropertyAge, TotalRooms, TotalBedrooms, AvgOccupancy, RoomsPerHousehold, BedroomsRatio.
- *Direct inputs:* NeighborhoodPop, Latitude, Longitude.
- Live updates are sent asynchronously to the backend on slider adjust, updating predicted prices dynamically with a neon green layout.

Section F: Critical Evaluation & Limitations (10 Marks)

1. Domain Suitability of the Model

- The Random Forest Regressor is highly suited for real estate valuation. It is robust to outliers (which are common in real estate due to luxury housing spikes) and captures the complex interactions between geographic coordinates and average occupancy.

2. Suitability of Deep Learning

- While Deep Learning (e.g. MLPs) can approximate the spatial price function, it is less suitable here due to:
- *Data scale:* Tabular dataset size (15k rows) is too small to train large network parameters without massive overfitting.
- *Interpretability:* Real estate appraisers require structural rules (e.g., how the bedrooms ratio impacts price) which are inspectable in tree splits but completely hidden in deep neural weights.

3. Limitations of the Current Approach

- **Temporal Shift:** The model does not include time metrics. Inflation, interest rates, and seasonal spikes are completely ignored.
- **Geographical Constraints:** The model is bound to the coordinate boundaries of the training dataset. It cannot predict prices for properties outside these geographical boundaries.

4. Future Suggestions

- We successfully implemented XGBoost and improved the R^2 score to **0.8256**. Future work could incorporate LightGBM to speed up training.
- Incorporate SHAP (Shapley Additive exPlanations) values to provide local feature contribution visualization in the web application UI.